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COMPARATIVE STUDIES OF CINKARA AND FERROUS GLUCONATE VITAMIN (FG) PREPARATION ON HEMATOLOGICAL AND BIOCHEMICAL PARAMETERS

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ABSTRACT

Cinkara is a herbal drug, used as a restorative and tonic by debilitated and convalescent patients, whereas FG is an allopathic medicine used to treat anemia. In present study the effects of these drugs have been evaluated on hematological and biochemical parameters. Cinkara is found to increase significantly the Hb concentration, erythrocyte count and platelet count and it significantly decreased the ESR, bleeding and clotting time. FG has significant effect on hematological parameters, except clotting time, osmotic fragility and fibrinogen. Among biochemical parameters FG has significantly influenced the levels of total protein and glucose. Cinkara also effected significantly the levels of blood glucose and total proteins.

INTRODUCTION

Anemia is most common public health problem, and those living in developing countries are at particular risk. Anemia is described as a decrease in the number of red blood cells per cubic mm or as a decrease in the hemoglobin concentration that is necessary for adequate tissue oxygenation (Keith, 1985).

The continuous replacement of mature blood cells, a process termed hematopoiesis, requires adequate supplies of minerals, both iron and copper, and a number of vitamins including folic acid, pyridoxine, ascorbic acid and riboflavin.

The drugs, which were compared in this study, contain a number of ingredients, and natural products probably with therapeutic potential. One of these drugs is FG, which is an allopathic hematinic used to treat anemia. Effects of this drug are compared with a herbal drug Cinkara, on hematological and biochemical parameters. As these parameters are of prime importance in detecting any abnormality in body systems.

"Cinkara" is a compound formulation prepared by Hamdard Laboratories (Waqf) Pakistan. According to the manufacturers, the product enriches blood, stimulates digestion and is an anti-infective. It is based on the extracts of some plants named as *Asparagus racemosus Willd., Aquilaria agallocha Roxb., Amomum subulatum, Cinnamomum cassiaa Blue, Cinnamomum tamala, Coriandrum sativum Linn., Crocus sativus, Cymbopogon jwarancusa Schult., Doronicum hookeri, Elettaria cardamomum, Ficus carica, Malus pumila Mill., Ocimum baslilicum, Parmelia perlata, Pimpinella anisum Linn., Pistacia vera Linn., Pterocarpus santalinus Linn., Punica grantum Linn., Prunus armeniaca Linn., Phoenix sylvestris Roxb., Rosa damascena Mill., Santalum album, Syzygium aromaticum Linn., Mentharvensis Linn.,* *Valeriana officianalis Linn., Zingiber officinale Roscoe and Onsoma bracteatum Wall (Medicinal herbal, 1996).*

Literature reveals that these plants are used as nutritive tonic, antibacterial, aphrodisiac, nerve tonic, carminative, refrigerant, antibilious, nutritional anemia prevention, febrifuge, in nausea and as repercussive. Chemical investigations of the plants reveal the presence of essential oils, saponins, quercetin, rutin, fixed oil, tannin, malic acid, colchicine, starch, sugar, insulin, coumarins, alkaloids, glycosides, albumin, calcium, potassium, magnesium, vitamin A, C, D, iron, phosphorus, chrysophanic acid, sucrose, free amino acids, choline, valeric acid (Chopra et al., 1956).

The allopathic drug used in this study contains (1) ferrous gluconate, (2) vitamin B1, (3) vitamin B2, (4) nicotinamide and (5) biotin. Present study is designed to observe the effects of drugs on some hematological and biochemical parameters.

MATERIALS AND METHODS

Two groups of 10 human volunteers were selected regardless of their age and sex, from low-income society and were tested to determine pretreatment values of hematological and biochemical parameters. The drugs were obtained from medical stores and distributed to the volunteers. They were advised to take the medicines for 3 weeks orally. The recommended doses of drugs were:

Dose of Cinkara	20 ml/day
Dose of FG	15 ml/day

After the period of 21 days volunteers were tested to determine the effects on following parameters:

1. Bleeding time, 2. Clotting time, 3. Erythrocyte count, 4. Hemoglobin percentage, 5. Leucocyte count 6. Platelet count, 7. Erythrocyte sedimentation rate, 8. Osmotic fragility, 9. Fibrinogen level, 10. Glucose concentration, 11. Total protein concentration, 12. Total lipid concentration.

1. Bleeding Time:

Bleeding time was measured by using Duke`s method which is based on measuring the time interval from inflicting a standard wound until the bleeding stops. (Duke, 1912; Adelson and Crosby, 1957; Heinrich, 1962).

2. Clotting Time:

Fingertip was punctured with a blood lancet and blood was taken up into a capillary tube, after every 30 seconds capillary was broken and appearance of clot was checked.

3. Erythrocyte Count:

For counting red blood corpuscles, blood is diluted with hayem`s reagent and red blood cells were counted in 80 small squares (Haug, 1959).

4. Hemoglobin Percentage:

Sahli`s method is used to estimate the concentration of hemoglobin in blood.

5. Leucocyte count:

Turk`s solution is used for diluting white blood cells. The counts were taken in 4 large squares of the neubauer chamber and counting was carried out under low magnification (about x 100).

6. Platelet count:

Ammonium oxalate solution was used and thrombocytes were counted in 4 large squares of Neubauer chamber (Brecher *et al.*, 1953).

7. Erythrocyte Sedimentation Rate (ESR):

Erythrocyte sedimentation rate is determined by Westergren method (Westergren 1922).

8. Osmotic Fragility:

The resistance of erythrocytes against a hypotonic sodium chloride solution was tested by preparing a dilution series.

9. Fibrinogen Level:

Fibrinogen level was determined by mixing blood and sodium citrate in 1:9 ratio, following centrifugation and supernatant was allowed to form clot with fibrinogen reagent (Koepke et al., 1975).

10. Glucose Concentration:

Glucose concentration was estimated on the basis of Trinder reaction (Trinder, 1969).

11. Total Protein Concentration:

Biuret method was used for the determination of total proteins in serum (Henry *et al.,* 1974).

12. Total Lipid Concentration:

Sulphophosphovanilline method was used for the determination of total lipids in serum (Zoeliner, 1962).

Statistical analysis:

All the parameters determined in normal and treated volunteers were subjected to the statistical analysis by applying paired t-Test.

RESULTS

Results are summarized in Table 1 (Cinkara) and Table 2 (FG Preparation).

Hematological Parameters:

Cinkara has significantly increased the Hb concentration ($P<0.01$) Fig. 4, erythrocyte count ($P<0.01$) Fig. 3, platelet count ($P<0.05$ (fig. 6) and significantly decreased the ESR ($P< 0.05$) Fig. 7, bleeding Fig. 1 and clotting time Fig. 2 ($P<0.01$).

Cinkara has caused insignificant decrease in leucocyte count (Fig. 5), and insignificant increase in fibrinogen (Fig. 9), No Significant change was noted in the osmotic fragility (Fig 8). FG has produced significant increase in the levels of hemoglobin, erythrocyte and platelet count, and it has produced significant decreases in the levels of leucocytes, ESR, bleeding time. No significant change was observed in fibrinogen, osmotic fragility and clotting time.

Biochemical Parameters:

Cinkara has significantly increased the glucose (Fig. 10) and total protein concentration (Fig. 11), Cinkara has no significant effect on lipid concentration (Fig. 12).

FG has caused significant increase in glucose (Fig. 10) and total protein (Fig. 11) but it has insignificantly increased the total lipid concentration (Fig. 12).

DISCUSSION

In the present study increase in hemoglobin and erythrocytes was seen by both drugs. This increase is because of iron content of these drugs. Iron has the capability to increase erythropoietic production (Sjaasted et al. 1996). Cinkara also contains minerals such as K, Ca, Mg which play an important role in erythropoiesis. Osmotic fragility remains unaffected because of vitamin A and C content of herbal drug. These vitamins maintain glutathione (antioxidant) and inhibit peroxidation in red blood cells. Vitamin C is an important antioxidant which is necessary for the synthesis of red blood cells (Behrman *et al.,* 1987). Vitamin C enhances absorption of iron, activates folic acid by reduction, improves erythropoiesis.

Ascorbic acid improves erythrocyte sedimentation rate (ESR) and hemoglobin. In the present study decrease in ESR is noted which is correlated with increased erythopoiesis. Increase in proteins is probably due to the presence of polyamines of Cinkara such as spermine and spermidine. These polyamines activate kinases involved in protein synthesis and increase RNA and DNA content. Fibrinogen is also increased which itself is a protein. Bleeding and clotting times are decreased which are correlated with increase in fibrinogen.

Cinkara has coumarins which inhibit platelet aggregation induced by adenosine. In present study platelets were significantly

Table-1
Effect of Cinkara Treatment on Hematological and Biochemical Parameters of Human Volunteers

	Normal value X ± S.D.	**Treated Value X ± S.D.**
Bleeding Time (Second)	224.5±45.5	218.9±45.3**
Clotting Time (Second)	334±49.3	330±49.3**
Erythrocyte count (M/cmm)	3.44±0.37	4.58±0.4**
Hemoglobin (g/dl)	10.62±1.24	13.58±1.1**
Leucocytes (/mm^3)	7490±2040.3	7140±1404.1
Platelets (/mm^3)	2,32,000±19465.0	2,50,000±21984.8*
E.S.R (mm/hr)	20.4±7.5	11.6±5.2*
Osmotic fragility (%)	0.41±0.012	0.41±0.008
Fibrinogen (mg/dl)	269.1±39.5	277.4±27.1
Glucose (mg/dl)	101.1±24.6	119.3±15.9**
Total protein (g/dl)	7.49±0.27	7.69±0.19**
Total lipid (mg/dl)	749.7±59.6	753±33.3

Data are expressed as mean ± S.D. Each figure is the mean of ten readings. Statistical analysis by paired t-test. *P<0.05, **P<0.01 as compared to normal.

Table-2
Effect of FG Treatment on Hematological and Biochemical Parameters of Human Volunteers

	Normal value X ± S.D.	**Treated Value X ± S.D.**
Bleeding Time (Second)	227.8±23.8	221.8±23.9*
Clotting Time (Second)	347.6±29.5	343.3±28.4
Erythrocyte count (M/cmm)	3.7±0.46	3.81±0.46*
Hemoglobin (g/dl)	11.2±1.43	11.45±1.47*
Leucocytes (/mm^3)	7050±2058.7	6660±1995.6*
Platelets (/mm^3)	233500±25932.6	253500±16840.7*
E.S.R (mm/hr)	25.9±9.91	16.4±7.16*
Osmotic fragility (%)	0.41±0.02	0.4±0.02
Fibrinogen (mg/dl)	260±40.7	272.3±35.5
Glucose (mg/dl)	105±32	114.6±26.0*
Total protein (g/dl)	7.48±0.29	7.65±0.26*
Total lipid (mg/dl)	704±116.6	732.5±57.8

Data are expressed as mean ± S.D. Each figure is the mean of ten readings. Statistical analysis by paired t-test. *P<0.05, **P<0.01 as compared to normal.

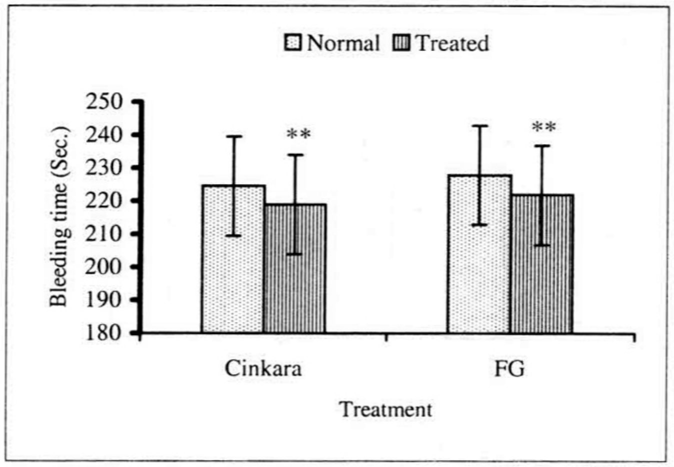

Fig. 1: Effect of treatment on bleeding time

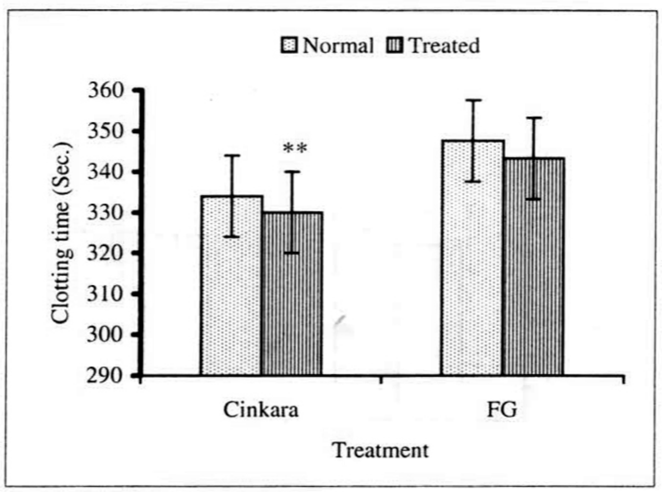

Fig. 2: Effect of treatment on clotting time

Vertical bars represent Mean ± S.E. (n=10) Statistical analysis was done using Student t-test. *P<0.05, **P<0.01

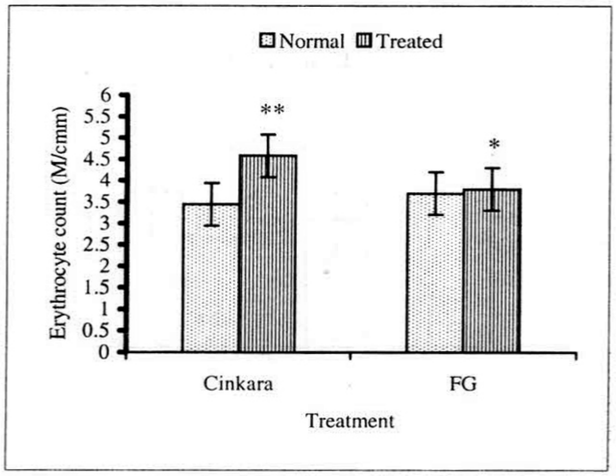

Fig. 3: Effect of treatment on erythrocytes

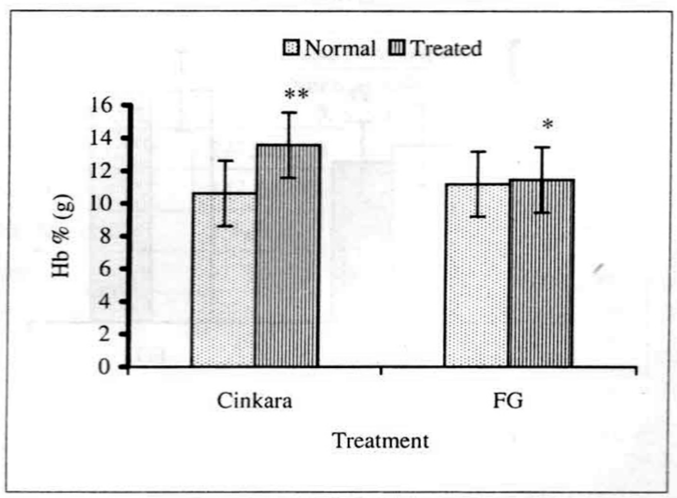

Fig. 4: Effect of treatment on hemoglobin

Vertical bars represent Mean ± S.E. (n=10) Statistical analysis was done using Student t-test. *P<0.05, **P<0.01

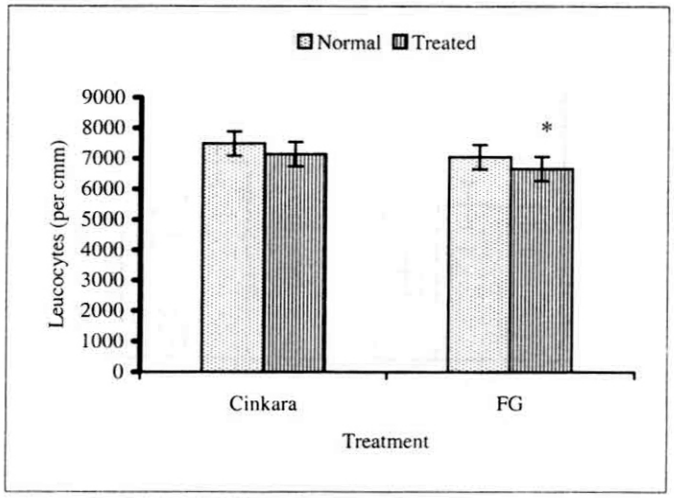

Fig. 5: Effect of treatment on leucocytes

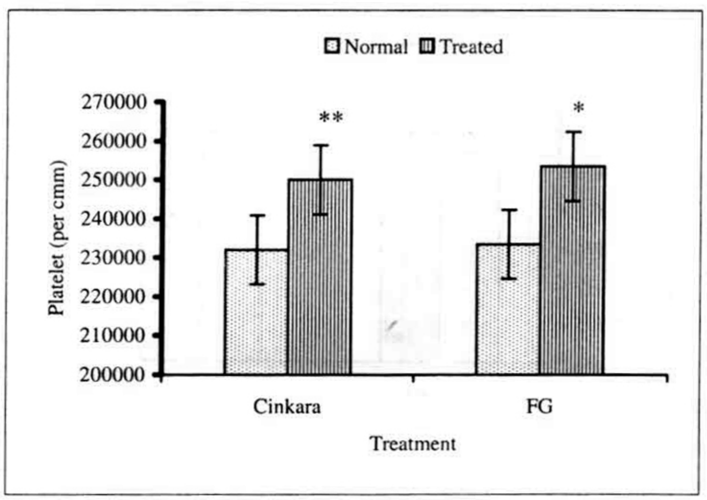

Fig. 6: Effect of treatment on platelets

Vertical bars represent Mean ± S.E. (n=10) Statistical analysis was done using Student t-test. *P<0.05, **P<0.01

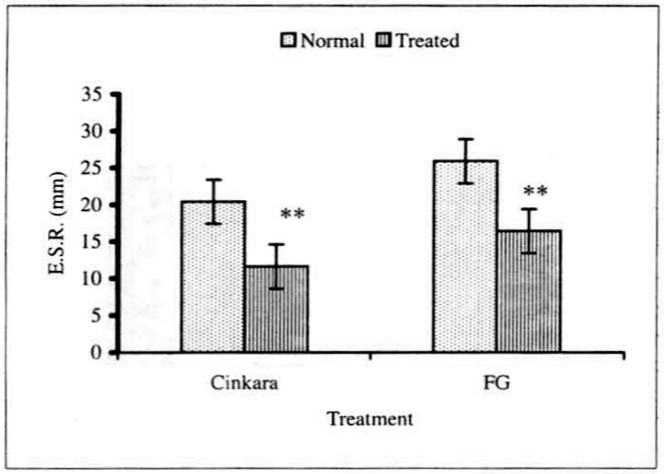

Fig. 7: Effect of treatment on E.S.R.

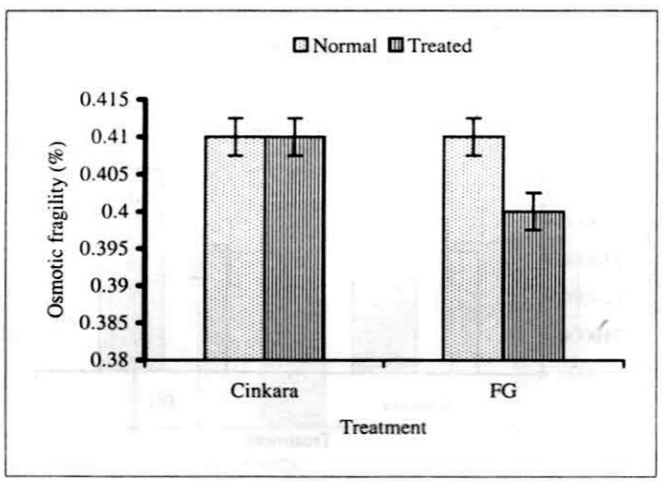

Fig. 8: Effect of treatment on osmotic fragility

Vertical bars represent Mean ± S.E. (n=10) Statistical analysis was done using Student t-test. *$P<0.05$, **$P<0.01$

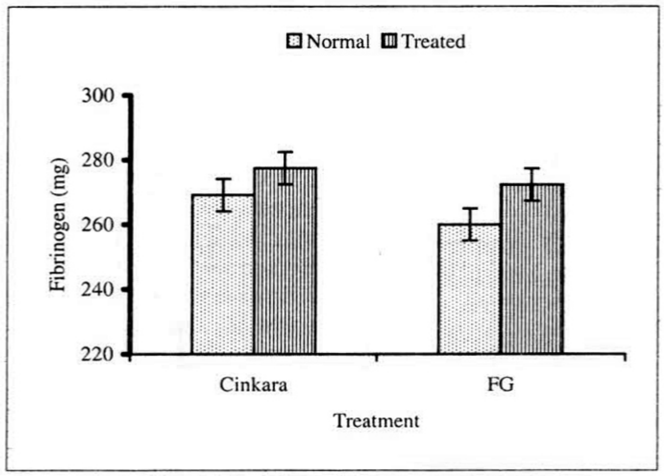

Fig. 9: Effect of treatment on fibrinogen

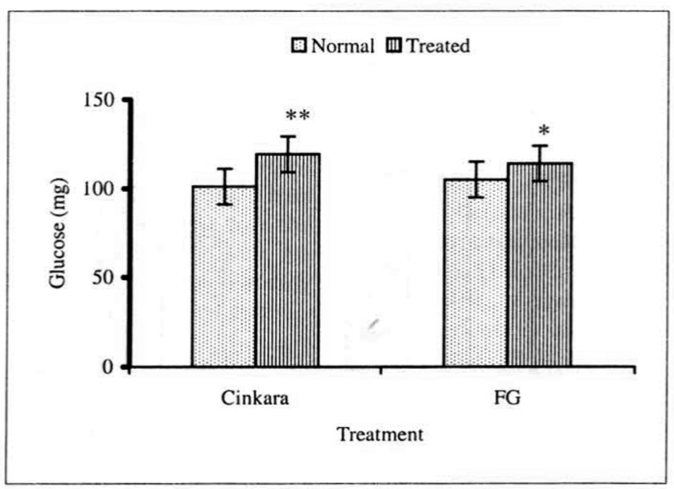

Fig.10: Effect of treatment on glucose

Vertical bars represent Mean ± S.E. (n=10) Statistical analysis was done using Student t-test. *P<0.05, **P<0.01

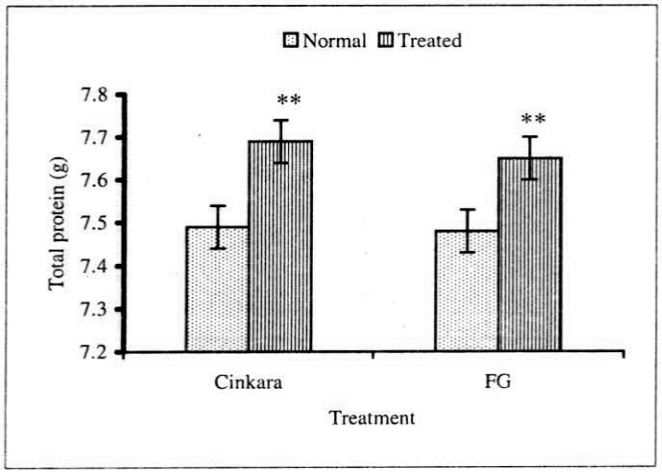

Fig. 11: Effect of treatment on total protein

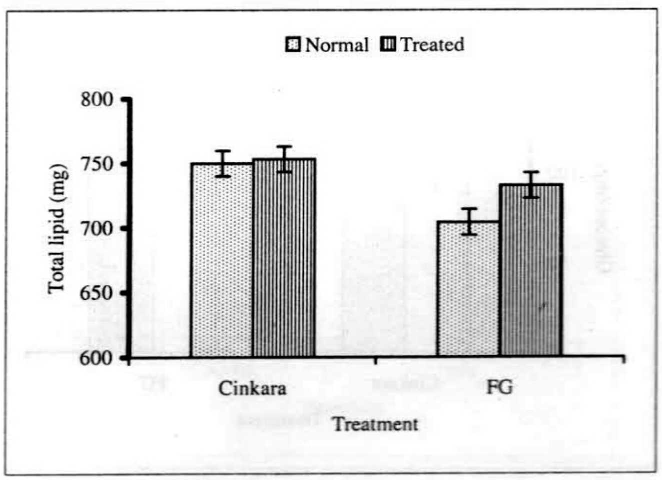

Fig. 12: Effect of treatment on total lipid

Vertical bars represent Mean ± S.E. (n=10) Statistical analysis was done using Student t-test. *P<0.05, **P<0.01

increased. Many growth factors have been reported in Cinkara, such as spermidine and spermine which are considered as growth factors in vitro and in cultured cells, Probably growth factors contribute to increased platelets.

Calcium which is a component of Cinkara regulates a number of enzymes such as glycogen synthetase, glycerol-3-phosphate dehydrogenase, pyruvate carboxylase, involved in affecting glucose concentration. This is correlated with increase in glucose concentration by Cinkara. Precursors of carbohydrate and components of TCA and Shikimate pathways of the drug may also be responsible for increase of glucose. Sterols may be related to increased steroid content and also useful as hypotensive agents.

FG contains ferrous gluconate, vitamin B1, B2, nicotinamide and biotin, Ferrous is involved in synthesis of hemoglobin and vitamin B1 is involved in metabolism of carbohydrate and many amino acids. This may be related to increased protein and glucose concentration. Biotin is a cofactor for enzymatic carboxylation of four substrates: (1) Pyruvate, (2) Acetyl Co A, (3) Propionyl CoA and (4) B-methyl crotonyl CoA. These reactions contribute to glucose formation and gluconeogenesis. Cinkara and FG both are erythropoietic. Effect on leucocytes, bleeding time and lipids required more detailed study.

Both medicines are scavengers of peroxidation, and are good erythropoietic agent and scavengers of hemolytic anemia. Drugs should be carefully evaluated in diabetic, hypercholesterolemic and patients of coagulation disorders.

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